

HUJI Applicative Research Report — Healthcare — October 2026

4 high-potential paper(s) · Healthcare · Monthly · October 2026 · Generated October 01, 2026

This month's digest highlights groundbreaking advancements across critical sectors, from novel drug delivery platforms to climate-resilient agriculture and targeted cancer immunotherapies. Researchers have developed a new method for intracellular delivery of genetic therapeutics, bypassing current logistical and cost barriers. Concurrently, a neuron-specific gene therapy framework offers a promising path for previously untreatable neurological disorders. Additionally, a significant agricultural innovation addresses global food security by enabling drought-resistant, high-yield wheat cultivars, alongside a new antibody therapy showing promise against difficult-to-treat colorectal cancers.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Rami I Aqeilan** — Drug Discovery, Neuroscience, Genomics, Synthetic Biology
- **Zvi Peleg** — AgriTech, Genomics, Synthetic Biology
- **Francesca Levi-Schaffer** — Drug Discovery, Immunology, Clinical

HEALTHCARE — RESEARCH HIGHLIGHTS

1. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake. 33/50

"Next-generation genetic therapeutics platform eliminates complex delivery barriers, unlocking broad intracellular targeting."

Main researcher: Eylon Yavin

The Institute for Drug Research, The School of Pharmacy, The Faculty of Medicine, The Hebrew University of Jerusalem, Hadassah Ein-Kerem, Jerusalem 9112102, Israel.

Published in ACS polymers Au · 2026-06

ABOUT THIS RESEARCH

Researchers developed a new type of synthetic DNA analogue called MCP PNA that can naturally enter cells without needing complex delivery vehicles. This solves a major historical bottleneck where these stable molecules were too large or poorly charged to cross cell membranes.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty

6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential

8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size

9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness

3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength

7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

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2. Neuron-specific WWOX gene therapy produces dose-dependent, durable rescue in a model of WWOX-related epileptic encephalopathy.

31/50

"Precision gene therapy platform delivers neuron-specific treatments for intractable neurological disorders."

Main researcher: Rami I Aqeilan

The Concern Foundation Laboratories, The Lautenberg Center for Immunology and Cancer Research, Department of Immunology and Cancer Research-IMRIC, Hebrew University-Hadassah Medical School, Jerusalem, Israel.

Published in Molecular therapy. Advances · 2026 Sep 10

ABOUT THIS RESEARCH

Researchers developed a targeted gene therapy that replaces the WWOX gene specifically in neurons to treat severe epilepsy and developmental disorders. The study demonstrates that this approach provides long-lasting and dose-dependent improvements in animal models of the disease.

COMMERCIAL ANGLE

The development of a neuron-specific delivery framework provides a platform technology for treating a wide range of neurodevelopmental genetic disorders currently lacking therapeutic options.

COMMERCIALISATION METRICS

Novelty

7/10

The research demonstrates significant translational innovation by moving beyond simple gene replacement to a 'rationally optimized' framework with neuron-specific targeting and dose-dependency. While gene replacement for WWOX is a logical therapeutic approach, the optimized delivery mechanism and the establishment of critical clinical parameters provide high-value, platform-enabling insights.

Commercial Potential

8/10

The study addresses a high-unmet-need orphan disease with a clearly defined therapeutic modality (gene replacement) and provides critical proof-of-concept regarding dose-dependency and durability. The use of a 'rationally optimized framework' suggests a potentially licensable platform technology for WWOX-related pathologies.

Market Size

4/10

The target market is limited to ultra-orphan populations (WOREE and SCAR12 syndromes), which represent extremely small patient cohorts. While the technology demonstrates high clinical utility, the addressable demand is constrained by the rarity of these specific genetic indications.

Tech Readiness

4/10

The research is at the preclinical stage, demonstrating efficacy and dose-dependency in a disease model rather than in humans. While it optimizes critical parameters for translation, it has not yet entered formal IND-enabling studies or clinical trials.

IP Strength

8/10

The development of a 'rationally optimized' neuron-specific framework suggests proprietary engineering of promoters or delivery vectors, which provides strong platform-level patentability beyond the transgene itself. This specificity enhances defensibility by creating a unique therapeutic window and reducing off-target risks compared to generic gene replacement.

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3. Antagonistic pleiotropy at the stem solidness 1 locus balances wheat stem architecture and yield components under terminal drought.31/50

"Climate-resilient wheat: novel genetic target boosts yield and drought resistance."

Main researcher: Zvi Peleg · zvi.peleg@mail.huji.ac.il.

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Published in TAG. Theoretical and applied genetics. Theoretische und angewandte Genetik · 2026-08

ABOUT THIS RESEARCH

Researchers identified a specific genetic locus (stem solidness 1) in wheat that controls the trade-off between stem structural strength and grain yield. The study explains how adjusting this genetic trait can help wheat plants survive terminal drought without sacrificing overall productivity.

COMMERCIAL ANGLE

Development of climate-resilient, high-yield wheat cultivars through precision breeding or CRISPR gene editing targeting the s1 locus.

COMMERCIALISATION METRICS

Novelty5/10

The identification of a specific locus controlling stem architecture and its trade-off with yield under drought is a valuable incremental contribution, but follows established paradigms of QTL mapping and antagonistic pleiotropy in crop science.

Commercial Potential7/10

The identification of a specific locus controlling the trade-off between stem strength and yield under drought provides a direct target for marker-assisted breeding or CRISPR editing in wheat, a global staple crop. While highly relevant for seed commercialization, the score is tempered by the inherent complexity of 'antagonistic pleiotropy' which may require precise tuning rather than a simple binary switch.

Market Size9/10

Wheat is one of the world's most critical staple crops; a genetic solution to maintain yield under terminal drought addresses a massive global food security market with high industrial demand.

Tech Readiness3/10

The research focuses on identifying a genetic locus and the biological trade-off (antagonistic pleiotropy) between stem architecture and yield, which is fundamental discovery research (TRL 2-3) rather than a developed seed product or breeding protocol ready for commercial deployment.

IP Strength7/10

The identification of a specific genetic locus (stem solidness 1) governing a trade-off between architecture and yield provides a clear target for Marker-Assisted Selection (MAS) or CRISPR editing, which is highly patentable in agricultural biotech.

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4. The anti-CEACAM1 agonistic antibody CCM5.01 inhibits the growth of colon cancer cell lines in 2D and 3D models.

30/50

"New antibody immunotherapy targets drug-resistant colorectal cancer, offering hope for MSS patients."

Main researcher: Francesca Levi-Schaffer · francescal@ekmd.huji.ac.il.

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Published · 2026 Aug 31

ABOUT THIS RESEARCH

Researchers have demonstrated that a specific monoclonal antibody, CCM5.01, can inhibit the growth of colorectal cancer cells by targeting the CEACAM1 receptor. The study shows that the antibody's effectiveness is linked to the ratio of long to short isoforms of the receptor, particularly in resistant tumor types.

COMMERCIAL ANGLE

Development of a novel immunotherapy targeting the CEACAM1 pathway to treat microsatellite-stable (MSS) colorectal cancers that are currently resistant to standard checkpoint inhibitors.

COMMERCIALISATION METRICS

Novelty

5/10

The research provides important validation of an existing lead candidate (CCM5.01) in a new disease indication (CRC) rather than introducing a novel mechanism or a new class of molecule. While the link between the L/S isoform ratio and efficacy is valuable for precision medicine, the core innovation (the agonistic antibody) was already established in prior work.

Commercial Potential

7/10

The work identifies a specific, actionable therapeutic target (CEACAM1) and a lead candidate (CCM5.01) for the high-value MSS colorectal cancer market, which is currently underserved by existing immunotherapies. However, the commercial potential is tempered by the fact that the data is limited to in vitro and 3D models, lacking the in vivo efficacy data required to validate a robust drug development program.

Market Size

9/10

The target addresses the massive and high-unmet-need colorectal cancer market, specifically the large microsatellite-stable (MSS) patient population that is currently resistant to standard immunotherapies. By targeting CEACAM1, this approach has the potential to unlock a multi-billion dollar market segment within oncology.

Tech Readiness

3/10

The research is currently in the preclinical, in vitro stage, focusing on cell line proliferation and 3D spheroid models. It lacks in vivo animal data or pharmacokinetic studies necessary to move toward clinical candidate status.

IP Strength

6/10

The innovation centers on a specific monoclonal antibody (CCM5.01) targeting a validated oncology target, which is highly patentable; however, the abstract focuses on mechanistic validation in cell lines rather than demonstrating a novel platform or proprietary structural modifications that would ensure high defensibility against future competitors.

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